

LayoutVLM: Differentiable Optimization of 3D Layout via Vision-Language Models

KAIST LAVA LAB
Hyeshim Kim

Contents

Overview	————	Positioning	<u>Preliminary</u>	Method	————	Validation	————	Discussion
				• Problem Formulation • Overview • Initialization • Differentiable Spatial constraint • Self-Consistent Decoding • Differentiable Optimization				

Overview

Paper info

Journal/Conference: **CVPR 2025**

Keywords: **Vision-Language Models (VLMs), Layout Generation**

Authors/Institution: **Sun et al. / Stanford, Google**



The key issue that the above paper focuses on

Semantic coherence

does the layout match the language instruction?

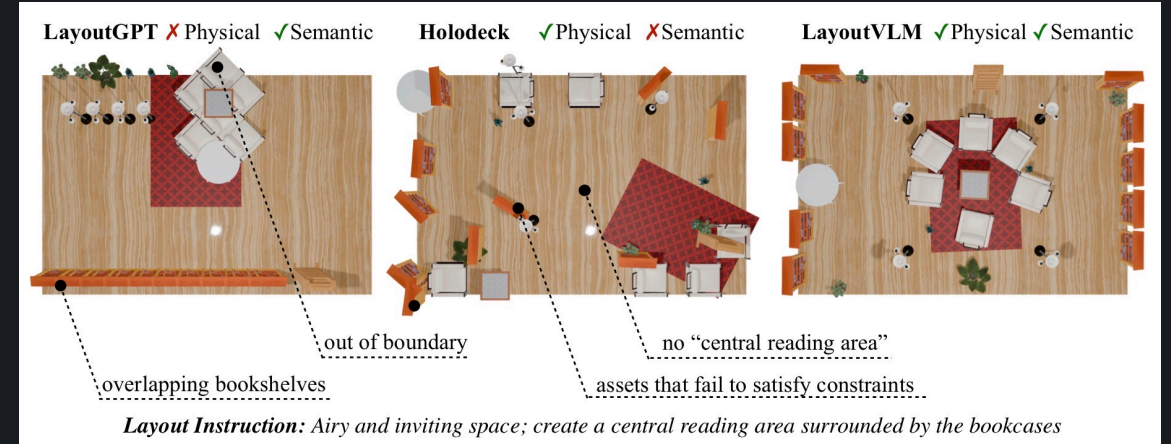
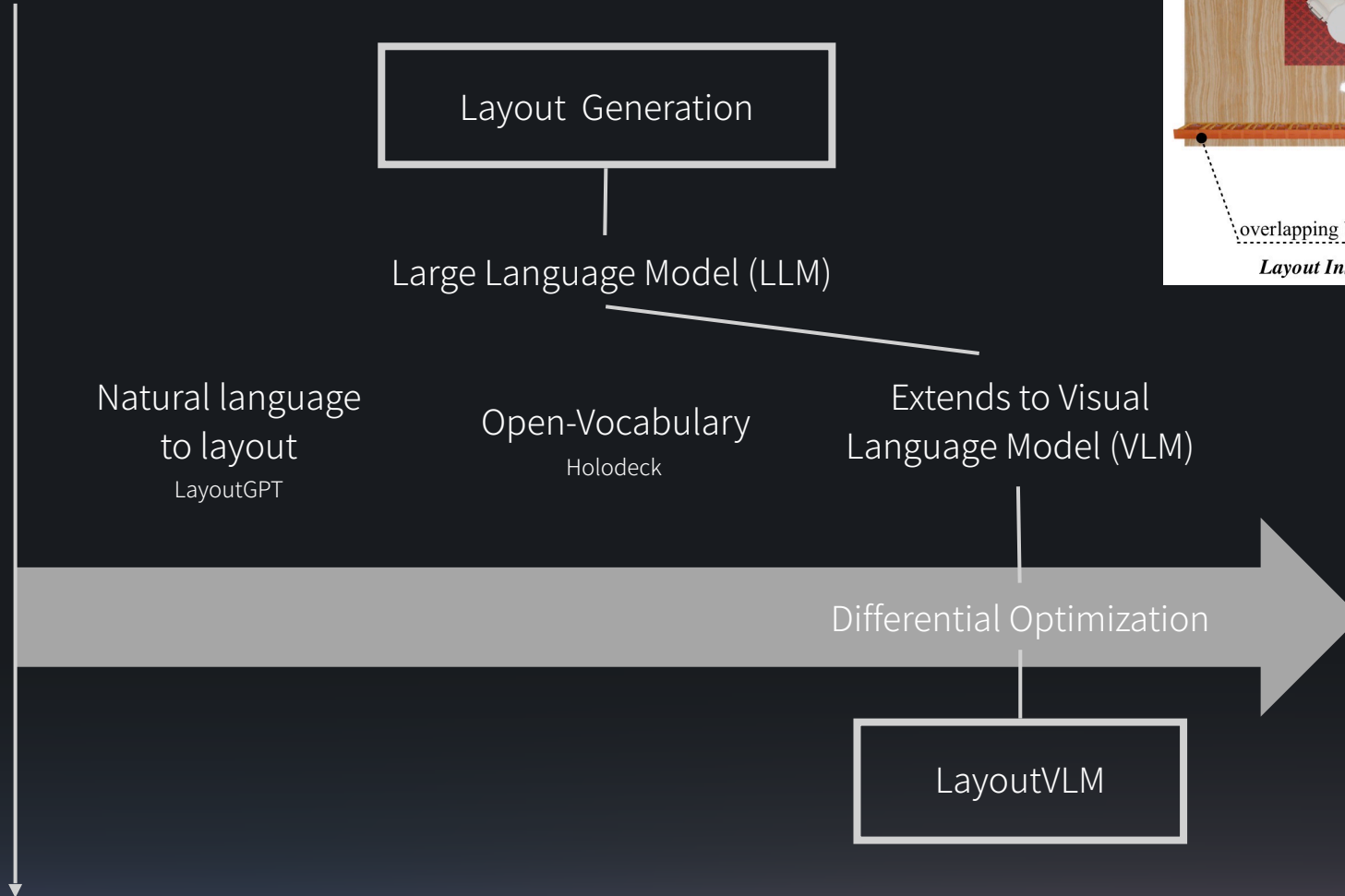
- A scene representation (poses + rules)
- Self-consistent decoding

Physical plausibility

are objects collision-free and inside the room?

- Differentiable optimization
- Projected gradient descent (PGD)

Positioning



- **Semantic coherence**
- **Physical plausibility**

Preliminary **Open-vocabulary**

Objaverse provides a huge, diverse library of 3D objects, enabling LayoutVLM to retrieve an asset for nearly any object described in language.



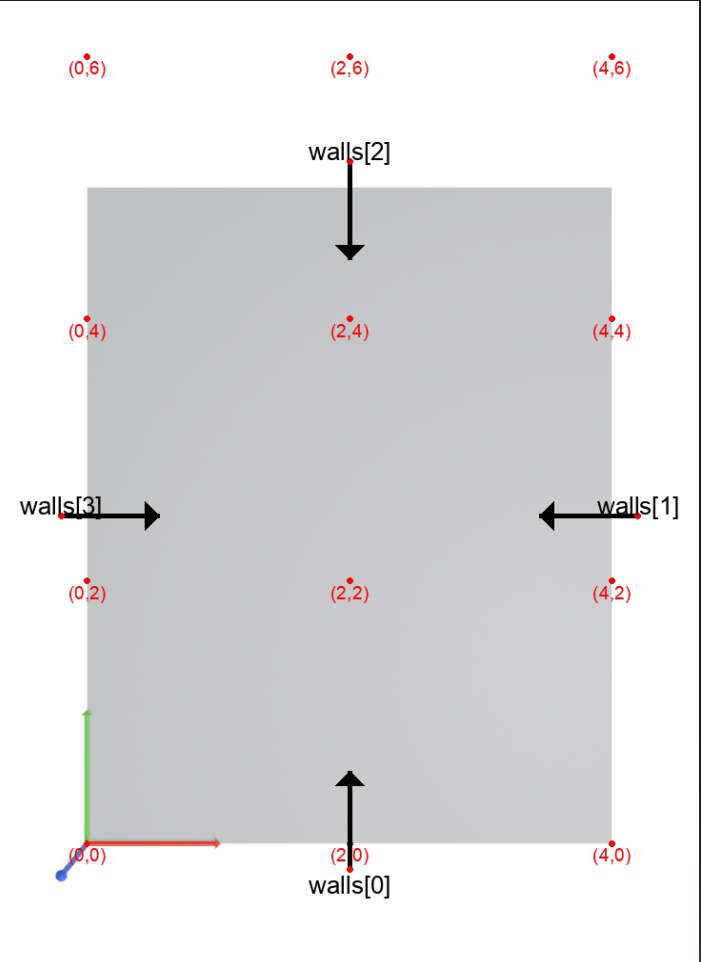
description: "This is a chair with a straight backrest featuring horizontal slats and a flat square seat, supported by four legs."

category: "chair"

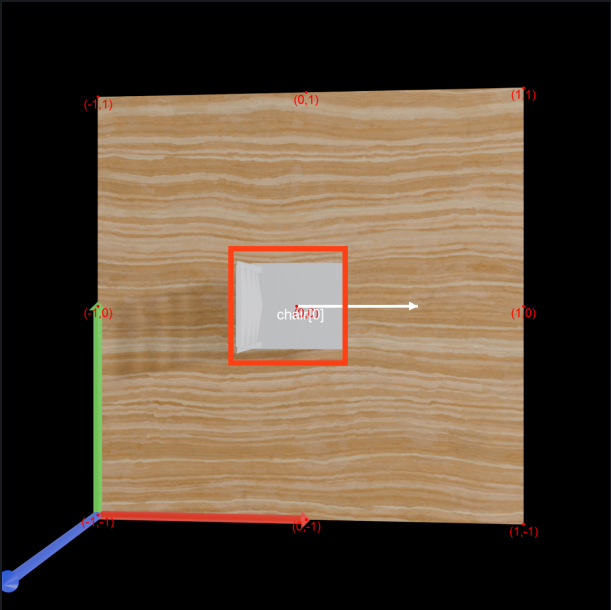
width: 45, **depth:** 45,
height: 90, **volume:** 60,

...

Method **Problem Formulation**



$$\{w_1, w_2, w_3, w_4\}$$



$$s_i$$

A short textual description of each object

$$b_i \in \mathbb{R}^3$$

the box's dimensions for each object
(width, depth, height)

$$l_{\text{layout}}$$

task_description: "a beach-inspired bedroom with light paint, a rattan chair and a queen bed, 4m x 5m"
layout_criteria: "airy, open, and serene with a focus on natural light and maximizing the feeling of space."
floor_vertices: $[[0, 0, 0], [4, 0, 0], [4, 5, 0], [0, 5, 0]]$
wall_height: 1.86
assets: "chair-0", "bed-0", ...



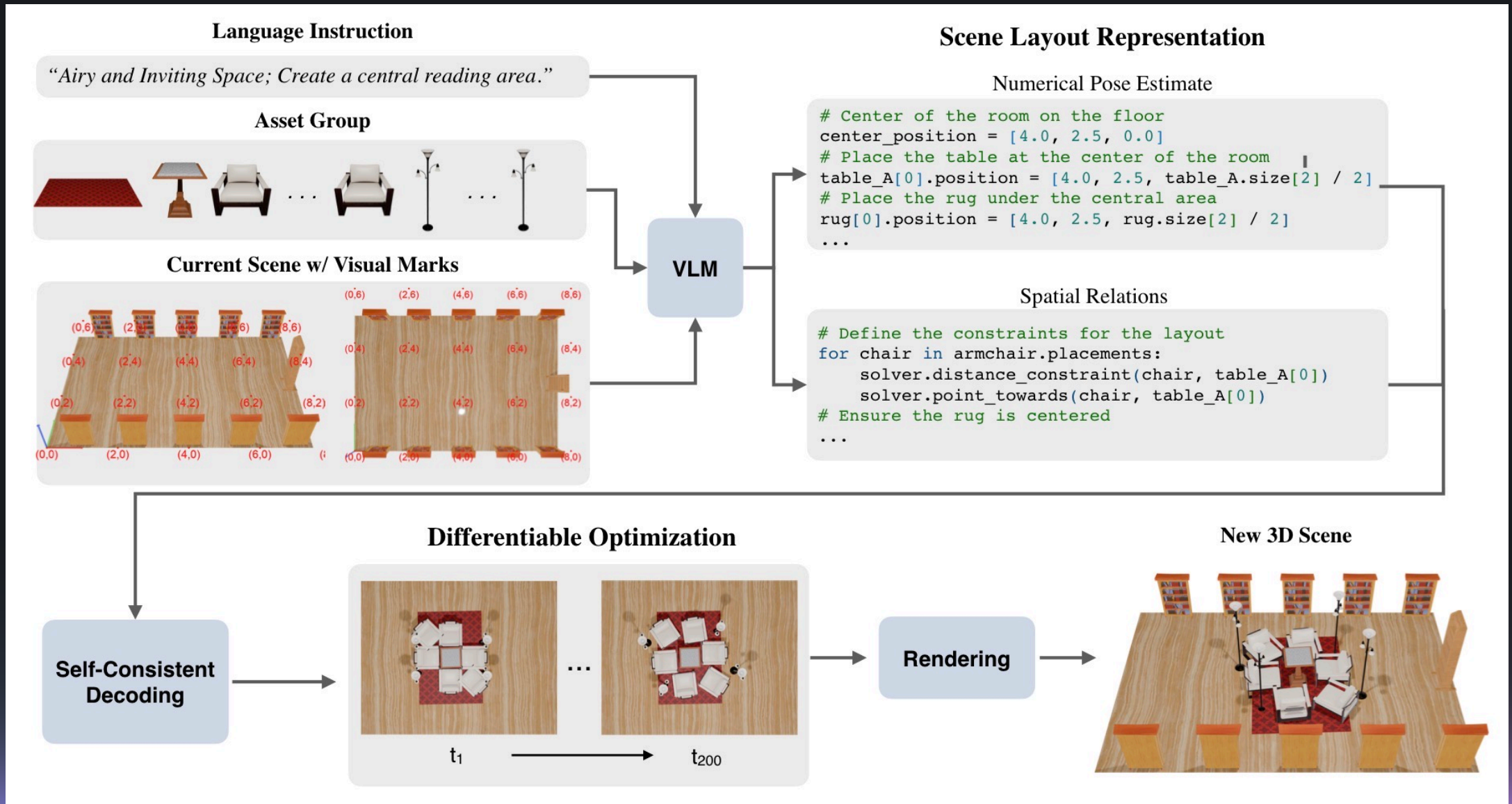
$$p_i = (x_i, y_i, z_i, \theta_i)$$

chair-0
position: $[2.5, 0.2, 0.4]$
rotation: $[0.0, 0.0, 89.9]$

bed-0
position: $[2.0, 4.0, 0.5]$
rotation: $[0.0, 0.0, -89.9]$

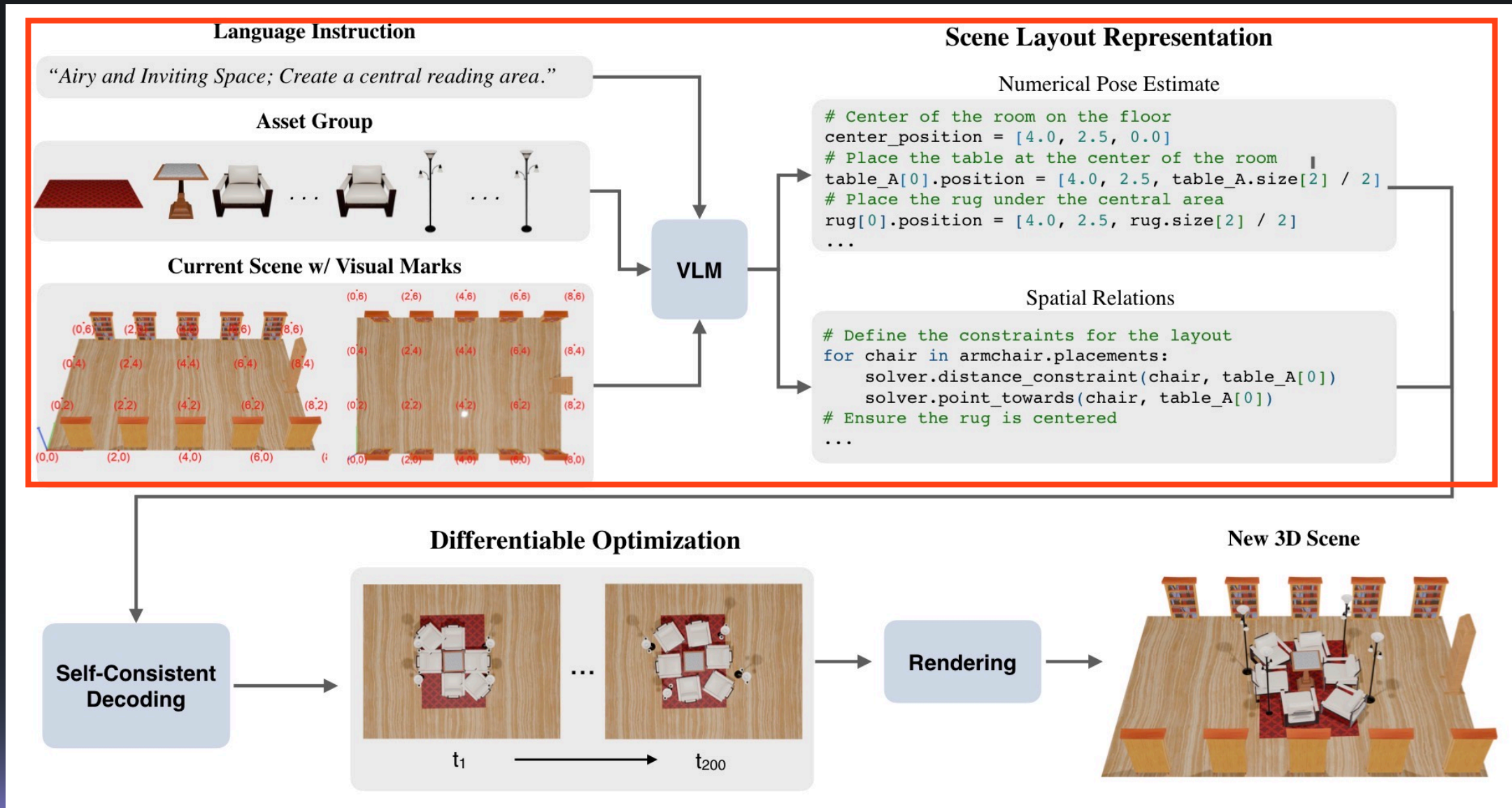
...

Method Overview



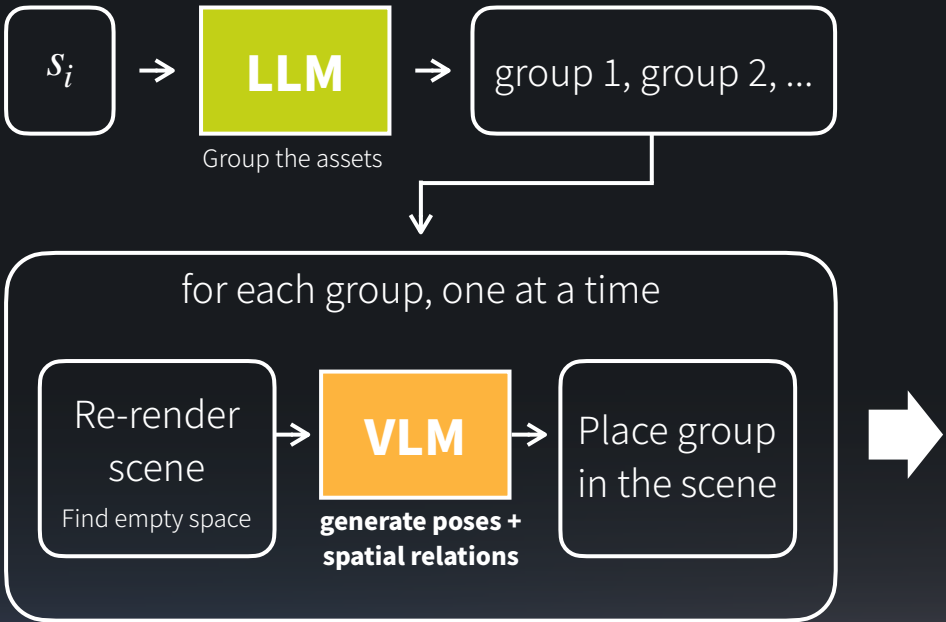
Method Overview

Initialization



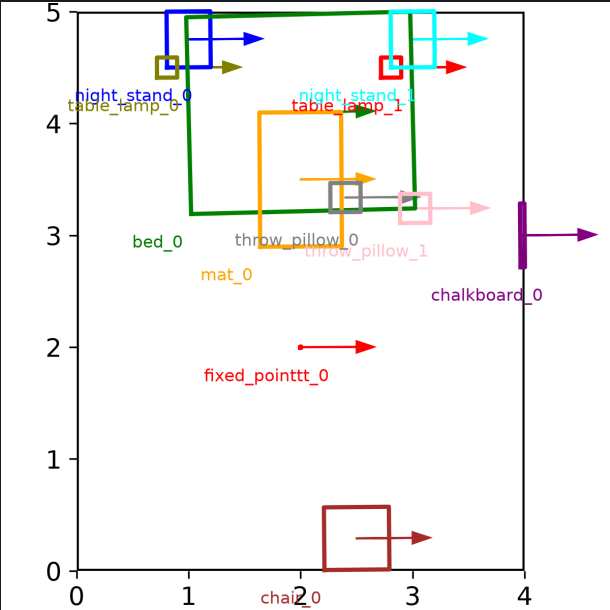
Method **Initialization**

VLMs are good at generating spatial relations for **pairs of objects**.
However, they struggle for **overall layout coherence**.



Visual Prompting

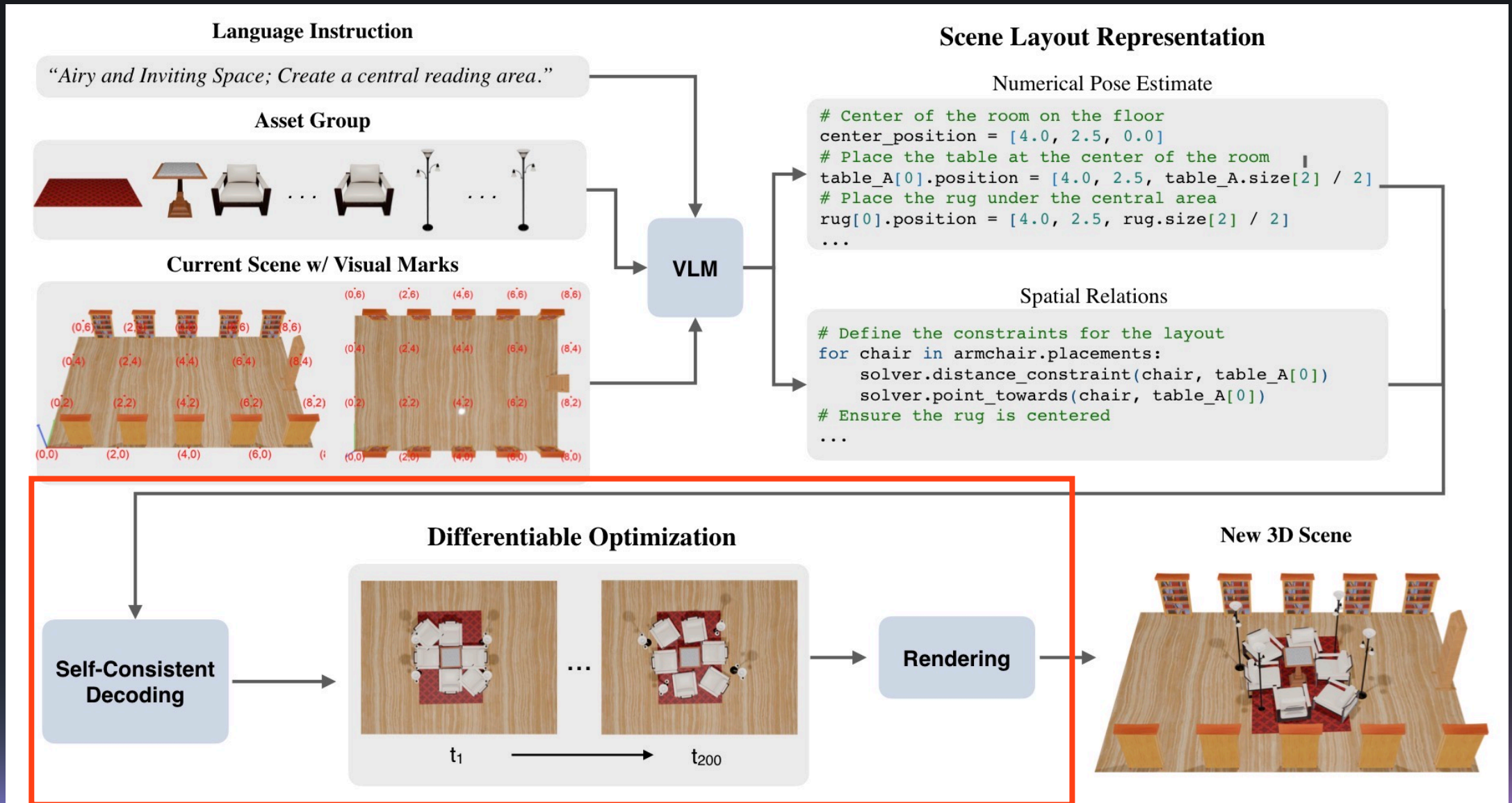
```
1 bed[0].position = [1.5, 3.1, 0.0]
2 bed[0].rotation = [0, 0, -1.57]
3 side_table[0].position = [0.3, 4.1, 0.0]
4 side_table[0].rotation = [0, 0, -1.57]
5 side_table[1].position = [2.7, 4.1, 0.0]
6 side_table[1].rotation = [0, 0, -1.57]
7 table_lamp[0].position = [0.3, 4.1, 0.5]
8 table_lamp[0].rotation = [0, 0, 1.57]
9 table_lamp[1].position = [2.7, 4.1, 0.5]
10 table_lamp[1].rotation = [0, 0, 1.5707]
11
12 for i in range(0, 2):
13     solver.against_wall(side_table[i], walls[2])
14     solver.distance(side_table[i],
15                     bed[0],
16                     min_distance=1.0,
17                     max_distance=3.0)
18 for i in range(0, 2):
19     solver.align_with(side_table[i], bed[0])
20 solver.point_towards(table_lamp[i], bed[0])
21 for i in range(0, 2):
22     solver.on_top_of(table_lamp[i], side_table[i])
```



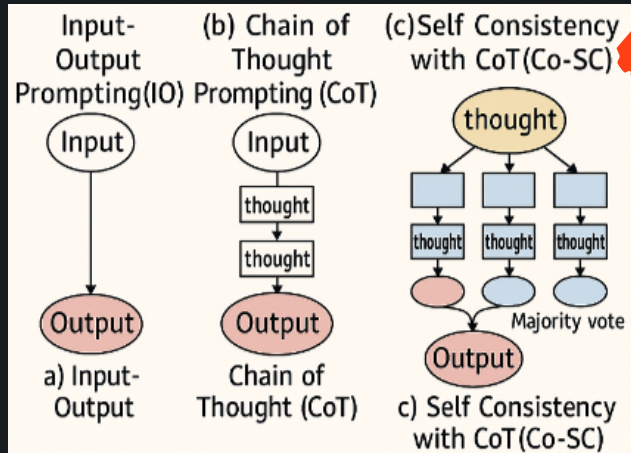
Type	Notation	Explanation
Positional	$\mathcal{L}_{\text{distance}}(p_i, p_j, d_{\min}, d_{\max})$	The distance between the two assets should fall within the range $[d_{\min}, d_{\max}]$.
Positional	$\mathcal{L}_{\text{on_top_of}}(p_i, p_j, b_i, b_j)$	Position one asset on top of another.
Rotational	$\mathcal{L}_{\text{align_with}}(p_i, p_j, \phi)$	Align two assets at a specified angle ϕ .
Rotational	$\mathcal{L}_{\text{point_towards}}(p_i, p_j, \phi)$	Orient one asset to face another with a offset angle ϕ .
Mix	$\mathcal{L}_{\text{against_wall}}(p_i, w_j, b_i)$	Place an asset again wall w_j .

Scene Layout Representation

Method Overview



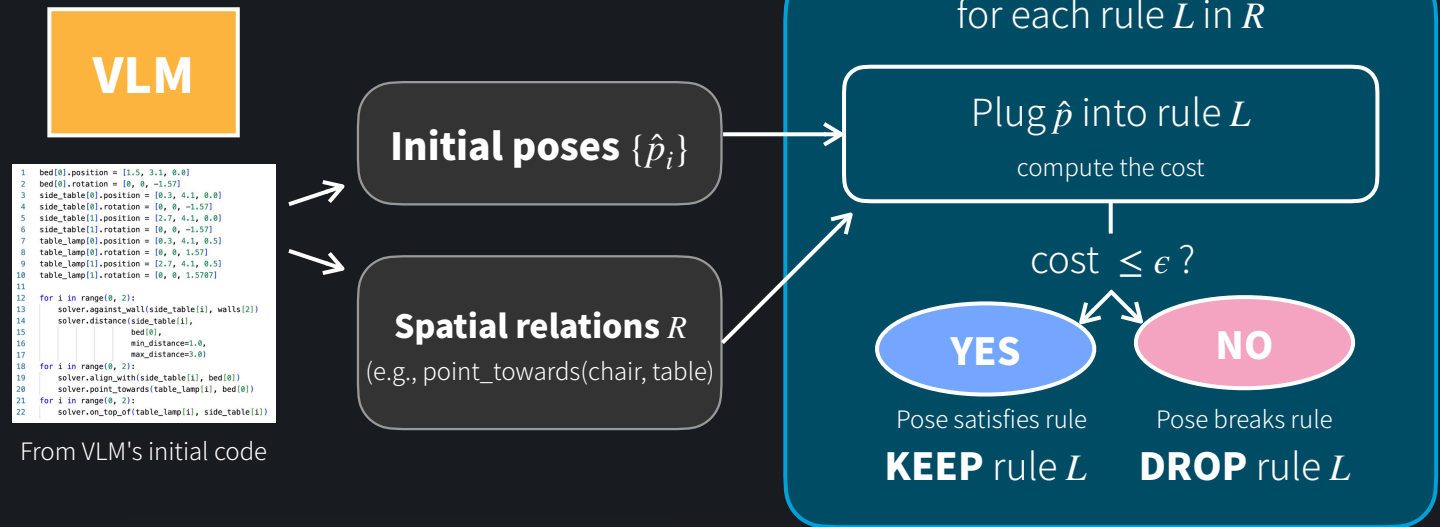
Method **Self-Consistent Decoding**



Standard Self-Consistency
in Prompt Engineering

Self-Consistency in LayoutVLM

Hypothesis: a rule is trustworthy when VLM's own numerical pose satisfies it. But if a rule contradicts the poses, it's likely noise.



$$\mathcal{L}_{\text{semantic}} = \sum_{L \in R} \mathbf{1} \left[L_i(\hat{p}_i, \hat{p}_j, \lambda) \leq \epsilon \right] \cdot L_i(p_i, p_j, \lambda)$$

Method **Differentiable Spatial constraint**

Type	Notation	Explanation
Positional	$\mathcal{L}_{\text{distance}}(p_i, p_j, d_{\min}, d_{\max})$	The distance between the two assets should fall within the range $[d_{\min}, d_{\max}]$.
Positional	$\mathcal{L}_{\text{on_top_of}}(p_i, p_j, b_i, b_j)$	Position one asset on top of another.
Rotational	$\mathcal{L}_{\text{align_with}}(p_i, p_j, \phi)$	Align two assets at a specified angle ϕ .
Rotational	$\mathcal{L}_{\text{point_towards}}(p_i, p_j, \phi)$	Orient one asset to face another with a offset angle ϕ .
Mix	$\mathcal{L}_{\text{against_wall}}(p_i, w_j, b_i)$	Place an asset again wall w_j .

Distance Objective: $\mathcal{L}_{\text{distance}}(p_i, p_j, d_{\min}, d_{\max}) = \text{clamp}\left(\min(\|p_i - p_j\| - d_{\min}, d_{\max} - \|p_i - p_j\|), 0, 1\right)$

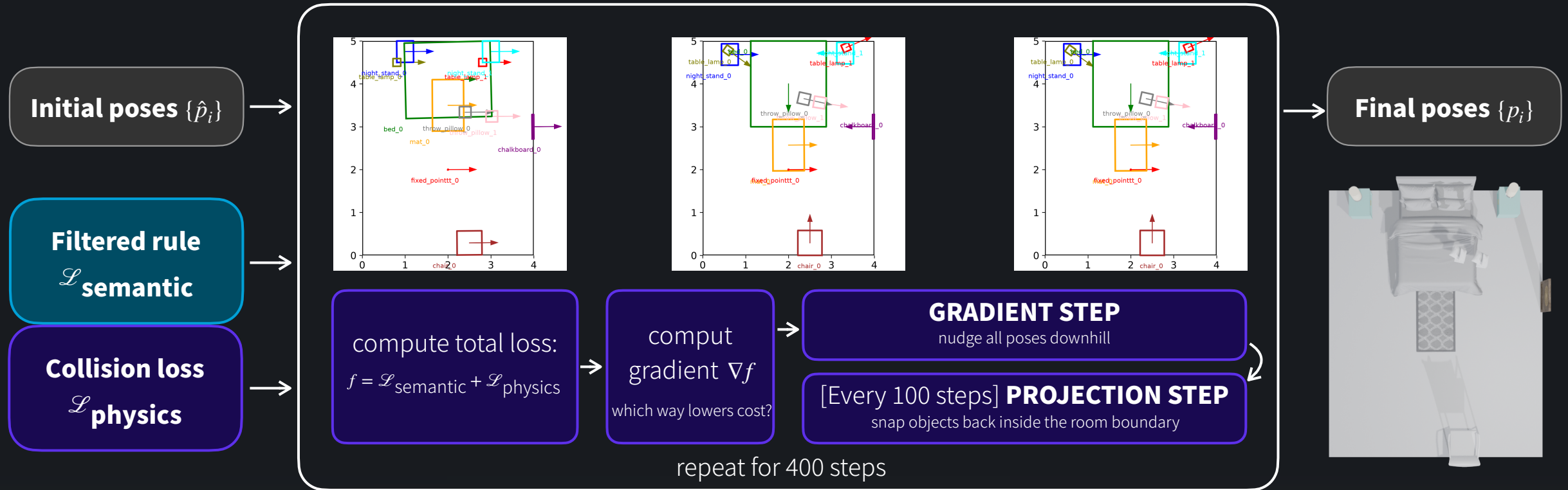
On-top-of Objective: $\mathcal{L}_{\text{on_top_of}}(p_i, p_j, b_i, b_j) = -\mathcal{L}_{\text{DIoU}}(p_i, p_j, b_i, b_j)$

Align-with Objective: $\mathcal{L}_{\text{align_with}}(p_i, p_j, \phi) = 1 - \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|}$ where $v_i = (\cos \theta_i, \sin \theta_i)$
Orientation vector

Point-towards Objective: $\mathcal{L}_{\text{point_towards}}(p_i, p_j, \phi) = \begin{cases} 0, & \text{if } v_i \cdot d_{ij} > 0, \\ 1 - \frac{v_i \cdot d_{ij}}{\|v_i\| \|d_{ij}\|}, & \text{otherwise,} \end{cases}$ where $d_{ij} = (x_j - x_i, y_j - y_i)$
the direction vector between the two objects' positions

Against-wall Objective: $\mathcal{L}_{\text{against_wall}}(p_i, w_j, b_i) = \sum_{k=1}^4 \text{clamp}(\|c_i^{(k)} - w_j\|, 0, 1) + \left(1 - \frac{v_i \cdot n_{w_j}}{\|v_i\| \|n_{w_j}\|}\right)$
where $c_i^{(k)}$ is the k-th corner of object i, and n_{w_j} is the normal vector of wall w_j .

Method **Differentiable Optimization**

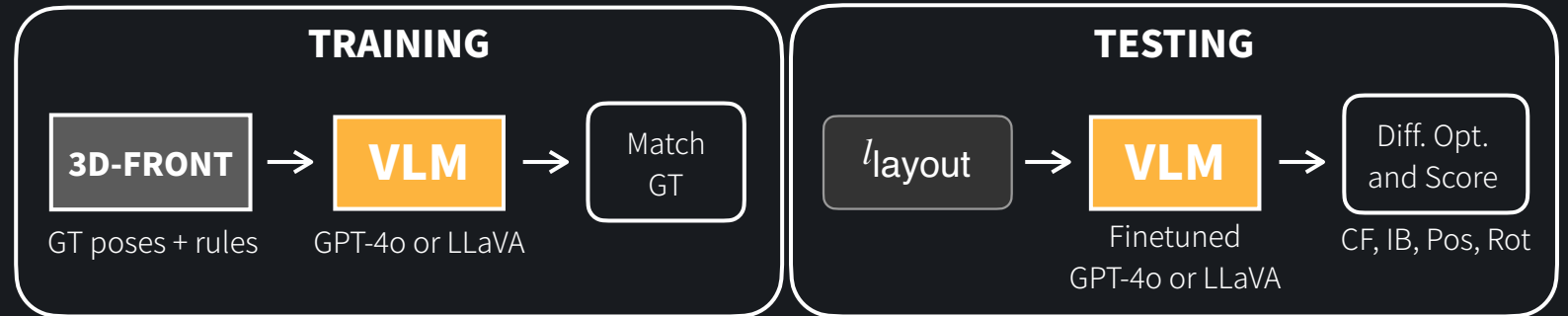


$$\mathcal{L}_{\text{physics}} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N L_{\text{DIOU}}(p_i, p_j, b_i, b_j)$$

IoU: Intersection over Union
 $\text{IoU} = \frac{\text{area of overlap}}{\text{area of union}}$
 no overlap=0, perfect overlap=1

Validation Finetuning VLMs

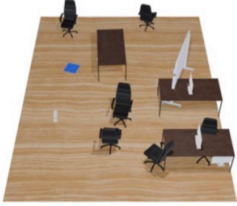
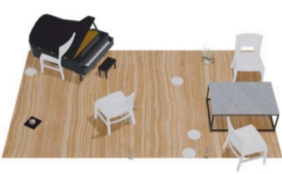
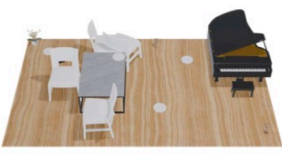
3D-FRONT Dataset
(rooms already arranged by humans (GT poses))



	Physics		Semantics		Overall
	Collision-Free Score CF	In-Boundary Score IB.	Pos.	Rot.	PSA
GPT-4o	66.7 ± 5.2	92.6 ± 3.0	71.4 ± 3.1	62.1 ± 2.4	43.2 ± 7.0
GPT-4o (FT on numerical)	77.8 ± 5.2	29.6 ± 3.0	73.8 ± 4.2	68.5 ± 1.9	11.9 ± 0.9
GPT-4o (FT on ours)	77.8 ± 3.0	96.3 ± 1.7	68.2 ± 0.9	62.5 ± 0.8	48.1 ± 1.8
LLaVA (random)	66.7 ± 0.0	3.7 ± 3.0	55.6 ± 1.7	47.5 ± 3.0	0.7 ± 0.6
LLaVA (FT on numerical)	77.8 ± 5.2	18.5 ± 3.0	68.4 ± 0.8	66.0 ± 1.3	6.8 ± 2.2
LLaVA (FT on ours)	85.2 ± 6.0	70.4 ± 8.0	73.8 ± 1.0	66.4 ± 0.5	39.5 ± 5.7

- open-source model (LLaVA) improves a lot
- fine-tuning on the scene representation > on raw numbers

Validation Benchmark Performance

	Florist Shop	Buffet Restaurant	Computer Room	Dining Room	Classroom
LayoutGPT					
Holodeck					
I-Design					
LayoutVLM					
Instructions	Unique and educational layout; position the large table centrally and succulents on top of it, surrounded by terrariums and potting stations	Compact yet functional layout; buffet tables arrangement should be symmetrically arranged	Professional and tech-integrated meeting space; center the conference table with computers, place the three tables in such a manner it forms one large longer table	Sleek and airy design; center the chairs and table, each chair should be equidistant from the table	Playful and flexible space; distribute colorful mats in a circular pattern with small tables interspersed, allowing for easy reconfiguration

Methods	Bedroom					Living Room				
	CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
LayoutGPT	100.0	66.7	85.7	85.9	52.2	44.4	11.1	74.7	64.4	9.6
Holodeck	88.9	22.2	69.3	67.9	14.1	77.8	0.0	66.3	55.6	0.0
I-Design	100.0	77.8	72.1	65.4	51.5	33.3	11.1	62.6	46.7	0.0
LAYOUTVLM	88.9	100.0	82.3	74.9	68.8	22.2	77.8	68.6	54.4	9.6

Methods	Classroom					Computer Room				
	CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
LayoutGPT	88.9	0.0	76.3	66.7	0.0	100.0	22.2	87.8	85.2	17.8
Holodeck	33.3	0.0	45.2	38.6	0.0	100.0	0.0	66.1	59.7	0.0
I-Design	55.6	11.1	50.7	47.0	0.0	88.9	22.2	74.0	70.7	8.9
LAYOUTVLM	77.8	100.0	74.6	68.6	48.3	100.0	88.9	85.4	84.5	77.0

Dining Room					Bookstore				
CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
88.9	22.2	76.0	68.9	14.8	88.9	55.6	80.9	79.4	35.9
88.9	0.0	38.0	36.6	0.0	55.6	0.0	65.7	59.0	0.0
88.9	66.7	76.4	66.4	34.8	66.7	11.1	68.1	69.4	5.2
88.9	100.0	63.4	56.9	51.1	55.6	100.0	82.0	82.8	49.8

Deli					Florist Shop				
CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
88.9	0.0	77.2	77.9	0.0	66.7	33.3	81.6	80.2	18.3
88.9	33.3	73.9	63.7	24.4	63.9	0.0	73.2	64.7	0.0
88.9	22.2	67.8	65.9	10.4	77.8	0.0	75.5	68.3	0.0
100.0	88.9	83.4	83.4	74.6	88.9	100.0	83.4	76.4	68.3

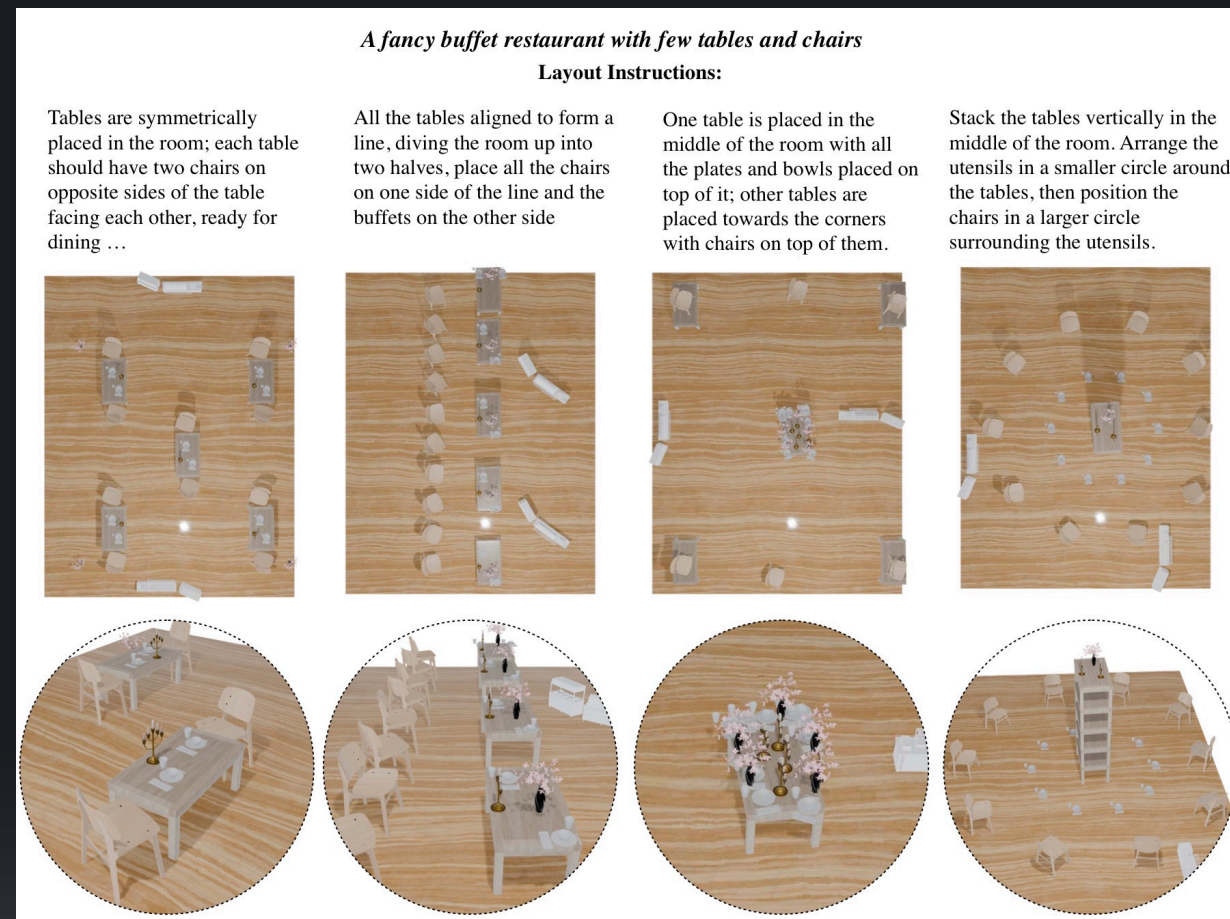
Buffet Restaurant					Children Room				
CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
100.0	33.3	81.2	83.3	26.9	100.0	0.0	80.9	82.6	0.0
77.8	11.1	47.7	42.4	7.4	77.8	22.2	72.7	70.0	18.7
100.0	55.6	63.5	57.1	35.2	77.8	55.6	78.1	75.1	34.8
88.9	88.9	74.3	64.8	51.5	100.0	100.0	81.9	88.2	88.5

Game Room					Average				
CF	IB	Pos.	Rot.	PSA	CF	IB	Pos.	Rot.	PSA
55.6	22.2	87.0	82.9	6.7	83.8	24.2	80.8	78.0	16.6
55.6	22.2	60.7	58.0	0.0	77.8	8.1	62.8	55.6	5.6
66.7	44.4	62.8	58.9	17.0	76.8	34.3	68.3	62.8	18.0
88.9	100.0	73.1	70.0	59.5	81.8	94.9	77.5	73.2	58.8

Validation Ablation Study

	Physics		Semantics		Overall Score
	CF	IB	Pos.	Rot.	PSA
LAYOUTVLM	81.8 ± 2.5	94.9 ± 2.2	77.5 ± 0.9	73.1 ± 1.1	58.8 ± 3.4
<i>Ablating the key components in LayoutVLM</i>					
w/o Visual Image	67.2 ± 1.1	94.9 ± 0.8	79.1 ± 1.3	74.0 ± 1.2	48.2 ± 1.5
w/o Self-Consistency	71.2 ± 2.5	92.9 ± 2.2	75.5 ± 1.1	72.1 ± 0.7	46.4 ± 2.4
w/o Const.	75.8 ± 2.5	14.1 ± 3.0	70.4 ± 1.3	67.3 ± 0.9	6.7 ± 1.6
<i>Ablating the two types of visual marks</i>					
w/o Visual Asset Mark	79.8 ± 0.6	87.9 ± 3.9	77.7 ± 0.6	72.2 ± 0.8	52.2 ± 5.0
w/o Visual Coordinate	77.2 ± 2.0	88.9 ± 1.7	74.3 ± 1.0	68.9 ± 1.3	46.0 ± 2.7
w/o Any Visual Mark	74.2 ± 6.4	84.9 ± 3.1	74.5 ± 0.8	69.2 ± 0.3	43.0 ± 2.2
<i>Ablating the proposed scene layout representation</i>					
w/o Numerical Init.	74.7 ± 4.0	86.6 ± 2.6	69.5 ± 0.1	65.4 ± 1.4	41.0 ± 1.7
w/o (Spatial) Const.	75.8 ± 2.5	14.1 ± 3.0	70.4 ± 1.3	67.3 ± 0.9	6.7 ± 1.6

- ① Placing objects from predicted poses alone, with no optimization, sends objects out of bounds and overlapping. This shows differentiable optimization is essential.
- ② Seeing rendered scenes and asset views helps the model plan spatially (e.g., arranging groups in distinct regions)



Discussion

Reasons for Acceptance of This Paper

- **Scene Representation:** A novel scene representation that improves both semantic coherence and physical plausibility.
- **Differentiable Optimization:** Reformulates layout generation as a differentiable optimization problem.
- **Thorough Evaluation:** Systematic evaluation across 11 room types, from benchmark comparison to ablation study, plus human validation.

Limitations of the paper

- **Quality:** Layouts still show failures (misalignment and overlapping objects appear frequently).
- **Requires Predefined Assets:** Asset retrieval is handled only in test-case setup, not integrated into the method itself
- **Limited Space Shapes:** Only rectangular 4-wall rooms are tested, so it likely handles only simple convex layouts

Classroom

"Playful and flexible space; distribute colorful mats in a circular pattern with small tables interspersed, allowing for easy reconfiguration."



Game room

"Intellectual and relaxing space; place the chess table near a wall, and the bookshelf along the opposite wall to balance the room."

